



OPEN Relying on AI at work reduces self-efficacy, ownership, and meaning while active collaboration mitigates the effects

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Artificial intelligence (AI) promises major productivity gains, but it also raises fundamental questions about how technology can reshape people's relationship to their work. Historical debates over industrialization warned that technological change could undermine people's connection to work and sense of meaning. Similar concerns now surround AI, where the key issue may not be *whether* AI is used, but *how* it is used. Across a pre-registered experiment (N = 269) and a follow-up survey (N = 270), we examine how different modes of AI use affect the confidence individuals have in completing work without AI assistance (self-efficacy), their sense of ownership over task output, and the meaning they perceive in their work. Participants completed occupation-specific writing tasks under one of three conditions: no AI use, passive AI use (copying AI-generated content), or active collaboration (drafting first and then using AI to refine). We find that passive use undermined self-efficacy, psychological ownership, and work meaningfulness, with declines in efficacy and meaningfulness persisting even when participants returned to manual work. In contrast, collaborative AI use preserved psychological connection to the task, producing outcomes comparable to independent work. Although passive use initially boosted enjoyment and satisfaction, these benefits reversed once participants resumed manual work. A complementary real-world survey mirrored these patterns across tasks beyond writing. Together, these findings show that the psychological consequences of AI use hinge on how it is integrated into human workflows, underscoring that strategies promoting active, collaborative use may help capture AI's productivity benefits while preserving human workers' agency, competence, and connection to their work.

Main

The purpose of life is not to be happy—but to matter, to be productive, to be useful, to have it make some difference that you lived at all.

— Leo Rosten, 1962

ChatGPT writes better than I do—and that's both amazing and deeply unsettling.

— Anonymous software engineer, Reddit AMA, 2023

Advancements in artificial intelligence are ushering in a new era, rapidly reshaping both the economic and psychological landscape^{1–5}. From a productivity perspective, generative AI has clear benefits, allowing both experts and novices to work at a faster pace and to access new skillsets^{6–8}. At the same time, the rapid deployment of AI usage has profound psychological implications: by changing the way that people access, synthesize, and produce content, AI has critical implications for how people learn and remember information, form relationships, and understand their place in the world^{9–17}.

Consideration of the human implications of AI^{18,19} are in line with a longstanding emphasis in the sociology and psychology of work on how new technologies reshape workers' subjective relationship to their labor. Classic perspectives emphasize that work is not merely a means of production, but a central source of human identity and meaning. Marx famously warned that when the products or processes of labor become increasingly external

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to the worker's control, work risks becoming alienating rather than fulfilling²⁰. Contemporary organizational scholarship has echoed this concern, emphasizing that meaningful work emerges when individuals experience a sense of agency, purpose, and connection to the outcomes of their effort^{21,22}. Three closely related constructs broadly capture these psychological experiences: self-efficacy, meaning of work, and psychological ownership.

Self-efficacy refers to individuals' beliefs about their capability to perform tasks and achieve desired outcomes through their own actions^{23,24}. In work contexts, self-efficacy is a critical psychological resource that shapes motivation, learning, and persistence, particularly when facing challenging tasks²⁵. Generative AI may undermine independent self-efficacy by assuming responsibility for tasks that workers previously performed. Even when performance improves, delegation of core tasks to AI may weaken workers' confidence in their own, non-AI enhanced abilities in that specific domain.

Work meaningfulness reflects the extent to which individuals perceive their work as purposeful, significant, and aligned with their values²¹. Meaning is fostered when individuals see a clear connection between their effort and valued outcomes, and when they experience themselves as authors of their work rather than mere executors²⁶. Generative AI complicates this process by obscuring the locus of contribution: when AI generates substantial portions of output, workers may struggle to perceive their own effort as causally linked to results. This ambiguity in authorship can diminish the sense of personal impact, a core pathway through which work becomes meaningful.

Psychological ownership captures the feeling that "this is mine" with respect to one's ideas and outputs, independent of formal ownership rights²⁷. Psychological ownership emerges through control over the work process, developing intimate knowledge through direct engagement, and investing personal effort. When outputs are co-produced with, or primarily generated by AI systems, all three pathways may be disrupted: workers exercise less direct control, possess less granular understanding of how outputs were created, and may perceive their personal investment as diluted. Consequently, workers may feel diminished attachment to results, even when they remain formally accountable for them.

These arguments suggest that generative AI has the potential to undermine multiple psychological foundations of work. Advancement in AI therefore risks creating a critical paradox: workers produce more, leading to greater wealth and advancement at the societal level, but at the expense of their own belief in themselves, experienced meaningfulness of their work, and sense of ownership of what they produce.

However, unlike machines in industrial settings, which are used in standardized ways, people use AI in *diverse* ways, even for the same task^{14,28}. Some adopt a passive approach, relying heavily on AI-generated content with minimal personal involvement. Others use AI more actively, retaining primary responsibility for content creation and using AI in a collaborative or supportive capacity. We thus also consider the critical question of whether the way in which AI is used influences its psychological consequences, asking whether active use of AI will lead to a different set of outcomes than passive use. To the degree that active, collaborative use can mitigate the negative psychological consequences of AI adoption, then the key to realizing its societal benefits may lie not just in *whether* AI is adopted, but in *how* it is integrated into human work.

Specifically, we argue that the extent to which AI erodes self-efficacy, meaningfulness, and psychological ownership depends critically on how responsibility and judgment are allocated between humans and machines. We draw on a growing literature in human–technology interaction to distinguish between augmentation where technology supports and extends human judgment, and automation where technology substitutes for human effort and decision-making^{1,29}. This distinction has primarily been used to explain differences in task allocation and productivity, but, critically, it may also impact how AI usage shapes workers' psychological experience of their work.

When AI augments human work, i.e., when workers use AI in an active way, they retain control over judgment, interpretation, and final decisions. This preserves the core pathways through which self-efficacy, meaningfulness, and ownership are built: workers still exercise agency, see their effort causally linked to outcomes, and invest themselves personally in results. In contrast, when AI automates or substitutes for human work, i.e., when workers use AI in a passive way, relying on AI with minimal personal involvement, these pathways are disrupted: Workers lose control over task execution, struggle to trace outcomes back to their own contributions, and invest less of themselves in outputs they did not substantially create. Thus, the same AI technology may generate fundamentally different psychological consequences, depending on whether it functions as a tool that amplifies human agency or as a replacement that bypasses it.

To examine these issues, we narrow our focus to writing tasks, an area where AI's productivity benefits are well documented^{30–32}. Following the approach of Noy and Zhang⁶, we recruited professionals from various roles and asked them to complete role-specific writing assignments. Consistent with our theoretical framework, we operationalize augmentation-oriented AI use as human-first, AI-supported collaboration, and automation-oriented use as passive reliance on AI-generated content. Specifically, we compared no AI use to two distinct AI usage modes (fully relying on AI to generate content versus using AI to edit human-written drafts), thereby capturing how varying levels of human involvement shape psychological outcomes.

Our focus was on examining how AI usage would influence core psychological outcomes that capture people's relationship to their work, including people's confidence in their ability to do the task on their own (AI-Independent self-efficacy)^{23,24}, sense of ownership of the produced outcome^{27,33}, and how meaningful they find the task^{21,34} (In the preregistration, we specified that we would measure both AI-independent and AI-dependent task self-efficacy. In the present paper, we treat AI-independent task self-efficacy as the focal outcome reported in the main text, as it aligns most directly with our theoretical interest in how AI use shapes individuals' perceived capability to perform tasks without AI assistance. Analyses of AI-dependent task self-efficacy are reported in the Supplementary Information). To provide additional potential insight, we further measured two evaluative responses tied to the task itself as exploratory outcome measures: task enjoyment (how pleasant the process was)^{35,36} and outcome satisfaction (how satisfied participants were with the final product)³⁷. We

also examined the durability of these potential psychological effects by assessing whether they persisted when participants subsequently completed a similar task without AI assistance. Taken together, this design allows us to move beyond documenting AI's productivity benefits to reveal how the manner of its use shapes workers' psychological experience of their work, both in the moment and in its lasting effects^{6,38,39}.

Finally, to complement the above experimental approach, we conducted a supplemental survey with a separate sample, examining work tasks beyond writing. This allowed us to assess whether the implications we consider extend to more varied, real-world uses of AI. While the survey does not permit causal inference, it provides broader insight into the extent to which our conclusions generalize across different types of tasks.

Results

Primary study

Outcomes following the primary task

We first report the main psychological outcomes (self-efficacy, psychological ownership, and meaningful work) measured immediately after participants completed the *Primary Task*. During this task, participants were randomly assigned to one of three AI usage conditions: *Copy and Paste AI* (using AI-generated content directly without modification), *First Human Then AI* (writing an initial draft manually and then using AI to refine it), or *No AI Use* (completing the task entirely without AI). Results are presented in Fig. 1.

Self-efficacy

A one-way analysis of variance (ANOVA) revealed a significant main effect of AI usage condition on AI-independent self-efficacy, $F(2, 266) = 3.54, p = 0.030, \eta^2_p = 0.026$. Pairwise comparisons using the Least Significant Difference (LSD) procedure indicated that participants in the *Copy and Paste AI* group ($M = 5.16, SD = 1.32$) reported significantly lower AI-independent self-efficacy than those in the *No AI Use* group, ($M = 5.63, SD = 1.38$), $p < 0.001$. The *First Human Then AI* group ($M = 5.43, SD = 1.17$) did not differ significantly from either the *No AI Use* group, $p = 0.355$, or the *Copy and Paste AI* group, $p = 0.239$.

Psychological ownership

Participants' reported sense of psychological ownership over their work also differed by AI usage condition, $F(2, 266) = 27.12, p < 0.001, \eta^2_p = 0.169$. LSD comparisons showed that participants in the *Copy and Paste AI* group ($M = 4.35, SD = 1.30$) reported significantly lower psychological ownership than both the *First Human Then AI* ($M = 5.26, SD = 0.78, p < 0.001$) and *No AI Use* groups ($M = 5.34, SD = 0.88, p < 0.001$). No significant difference emerged between the *First Human Then AI* and *No AI Use* groups ($p = 0.681$).

Meaningful work

Perceived meaningfulness of the task also varied significantly by AI usage condition, $F(2, 266) = 5.26, p = 0.006, \eta^2_p = 0.038$. LSD comparisons revealed that participants in the *Copy and Paste AI* group ($M = 4.94, SD = 1.59$) rated their work as significantly less meaningful than those in both the *No AI Use* group ($M = 5.54, SD = 1.24, p = 0.002$), and the *First Human Then AI* group ($M = 5.46, SD = 1.42, p = 0.034$). The *First Human Then AI* and *No AI Use* groups did not differ from each other ($p = 0.746$).

Outcomes following the subsequent manual task

Following the *Primary Task*, all participants completed a second writing task *without* any AI assistance, regardless of their original condition. In this section, we report psychological outcomes assessed immediately after the *Subsequent Manual Task*, presented by participants' AI usage condition during the *Primary Task* (see Fig. 2). This allows us to examine whether prior AI use had lasting effects on participants' psychological experiences, even after they returned to working independently without it. (As a supplemental analysis, we also conducted 3 [AI usage condition] $\times$ 2 [measure time point] mixed ANOVAs to consider the data as a whole across the two time points; the interaction results are reported in the Supplementary Information).

Self-efficacy

We conducted a one-way ANOVA on AI-independent self-efficacy measured after the *Subsequent Manual Task*. The analysis revealed significant lingering differences, $F(2, 266) = 6.25, p = 0.002, \eta^2_p = 0.045$. LSD tests revealed that participants who were in the *Copy and Paste AI* condition for the *Primary Task* reported lower AI-independent self-efficacy ($M = 4.95, SD = 1.68$) than those in the *First Human Then AI* condition ($M = 5.45, SD = 1.22, p = 0.055$) and the *No AI Use* condition ($M = 5.66, SD = 1.42, p < 0.001$). The *First Human Then AI* and *No AI Use* groups did not differ from each other ($p = 0.400$). These findings highlight that the negative impact of passive AI use on individuals' confidence may endure, persisting even after they resume working without AI.

Psychological ownership

A one-way ANOVA on psychological ownership measured after the *Subsequent Manual Task* revealed no significant differences ($p's > 0.344$). This indicates that the lower sense of ownership reported by the *Copy and Paste AI* group after the *Primary Task* rebounded after they completed a subsequent task on their own. These results suggest that psychological ownership may be closely tied to how the task is performed: when individuals produce work themselves, rather than relying on AI, their sense of ownership is restored.

Meaningful work

A one-way ANOVA on meaningfulness measured after the *Subsequent Manual Task* revealed significant lingering differences, $F(2, 266) = 3.40, p = 0.035, \eta^2_p = 0.025$. LSD tests indicated that participants who were in the *Copy and Paste AI* condition for the *Primary Task* reported lower levels of meaningfulness ($M = 5.07, SD = 1.62$)

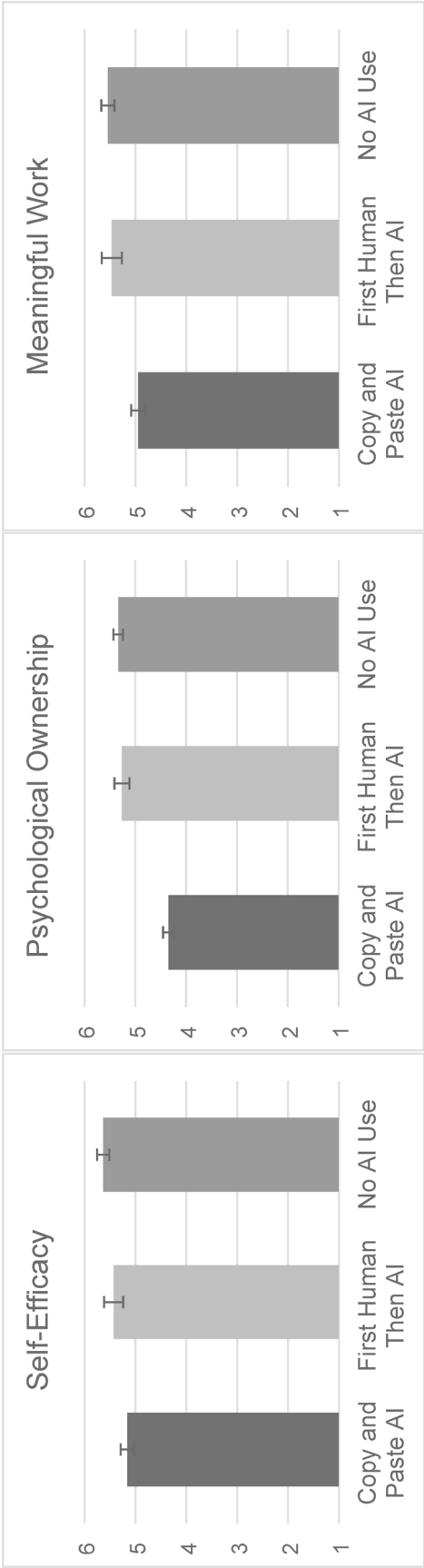


Fig. 1. Psychological outcomes following the *Primary Task*. Self-Efficacy, psychological ownership, and meaningful work immediately after participants completed the *Primary Task* across three AI usage conditions: *Copy and Paste AI* (using AI-generated content directly without modification), *First Human Then AI* (writing an initial draft manually and then using AI to refine it), or *No AI Use* (completing the task entirely without AI). Error bars represent SE.

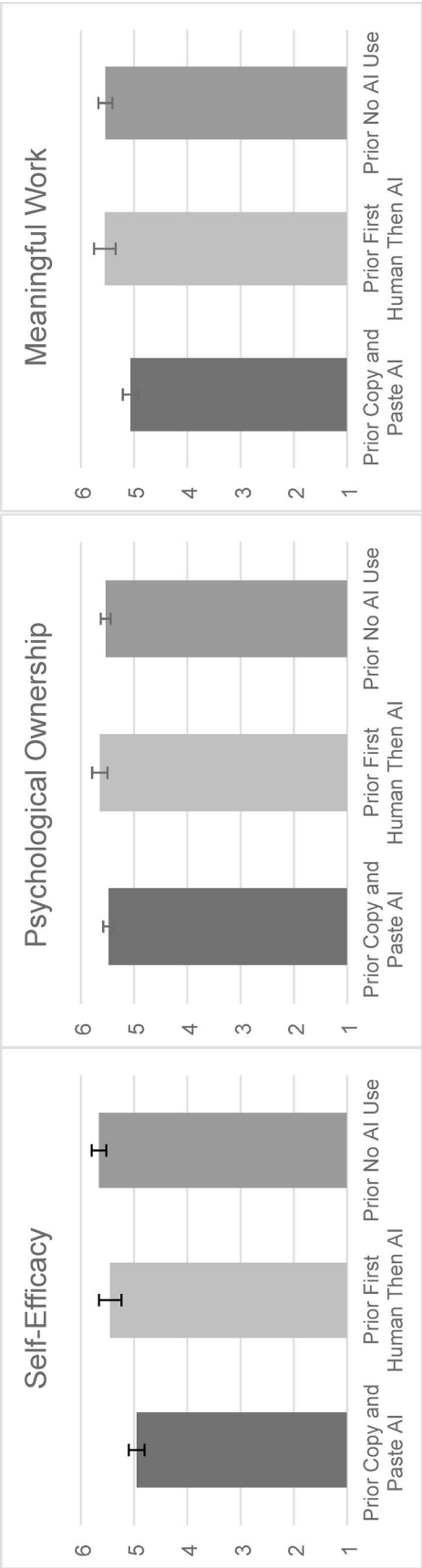


Fig. 2. Psychological outcomes following the *Subsequent Manual Task*. Self-efficacy, psychological ownership, and meaningful work measured after all participants completed the *Subsequent Manual Task* without AI assistance, shown by their prior AI usage condition in the *Primary Task* (*Prior Copy and Paste AI*, *Prior First Human Then AI*, *Prior First AI Then Human*, *Prior No AI Use*). Error bars represent SE.

than those in the *First Human Then AI* condition ($M = 5.55$, $SD = 1.34$, $p = 0.054$) and the *No AI Use* condition ($M = 5.54$, $SD = 1.31$, $p = 0.017$). No significant difference emerged between the *First Human Then AI* and *No AI Use* groups ($p = 0.961$). These findings suggest that passive AI use may continue to dampen individuals' sense of purpose and value in subsequent independent work.

Secondary dependent variables: task enjoyment and outcome satisfaction

Secondary outcomes following the primary task

Beyond these primary psychological outcomes, we also explored how AI use shaped secondary experiences of task enjoyment and satisfaction. As shown in Fig. 3, participants' reported sense of task enjoyment differed by AI usage condition, $F(2, 266) = 2.72$, $p = 0.068$, $\eta^2_p = 0.020$. LSD tests indicated that participants in the *Copy and Paste AI* group reported significantly higher task enjoyment ($M = 5.66$, $SD = 1.40$) than those in the *No AI Use* group ($M = 5.19$, $SD = 1.45$), $p = 0.020$. However, there were no statistically significant differences between the *First Human Then AI* group ($M = 5.42$, $SD = 1.67$) and either the *Copy and Paste AI* group ($p = 0.349$) or the *No AI Use* group ($p = 0.363$).

Participants' satisfaction with the outcome of the task also differed by AI usage condition, $F(2, 266) = 19.64$, $p < 0.001$, $\eta^2_p = 0.129$. Participants in the *Copy and Paste AI* group reported significantly higher satisfaction ($M = 5.80$, $SD = 1.20$) than those in both the *No AI Use* group ($M = 4.50$, $SD = 1.74$; $p < 0.001$) and the *First Human Then AI* group ($M = 5.26$, $SD = 1.61$; $p = 0.043$). Participants in the *First Human Then AI* group also reported significantly higher satisfaction than those in the *No AI Use* group ($p = 0.004$). Together, these results suggest that passive AI use may boost immediate task enjoyment and satisfaction with work outcomes.

Secondary outcomes following the subsequent manual task

A significant effect of AI usage condition during the *Primary Task* emerged on participants' task enjoyment following the *Subsequent Manual Task*, $F(2, 266) = 6.66$, $p = 0.002$, $\eta^2_p = 0.048$ (see Fig. 4). Participants who were in the *Copy and Paste AI* condition during the *Primary Task* reported significantly lower enjoyment ($M = 4.67$, $SD = 1.90$) than those in the *First Human Then AI* ($M = 5.56$, $SD = 1.55$, $p = 0.003$) and *No AI Use* ($M = 5.39$, $SD = 1.59$, $p = 0.002$) conditions. No significant difference was found between the *First Human Then AI* and *No AI Use* groups ($p = 0.566$).

Outcome satisfaction after the *Subsequent Manual Task* also revealed a significant effect of condition, $F(2, 266) = 10.98$, $p < 0.001$, partial $\eta^2_p = 0.076$. Participants who had previously relied on *Copy and Paste AI* reported lower satisfaction ($M = 4.12$, $SD = 2.02$) compared to those previously in the *First Human Then AI* ($M = 5.42$, $SD = 1.61$, $p < 0.001$) and *No AI Use* ($M = 4.98$, $SD = 1.59$, $p < 0.001$) conditions. No significant difference was found between the *First Human Then AI* and *No AI Use* groups ($p = 0.144$).

These results suggest that while passive AI use may enhance immediate enjoyment and satisfaction, this effect is short-lived and reverses when individuals subsequently perform the task manually, possibly due to contrast with the ease and quality of AI-generated output, which makes manual work feel more effortful and less engaging. In contrast, individuals who previously collaborated with AI or worked without AI showed more stable enjoyment and satisfaction, even after returning to manual work.

Follow-up correlational study

To complement our experimental investigation and explore how these dynamics manifest in real-world contexts, we conducted a follow-up correlational study. This study surveyed working adults who varied in their use of generative AI for professional tasks, distinguishing between passive reliance on AI (e.g., using AI to fully generate content) and active collaboration with AI (e.g., using AI to refine self-generated content). The results are broadly consistent with our experimental findings (see Table 1): passive reliance on AI was negatively correlated with self-efficacy ($r = -0.45$, $p < 0.001$), psychological ownership ($r = -0.22$, $p < 0.001$), and outcome satisfaction ($r = -0.16$, $p = 0.008$), but was not significantly related to work meaningfulness ($r = 0.05$, $p = 0.392$) or task enjoyment ($r = 0.09$, $p = 0.134$). In contrast, active AI collaboration was positively correlated with self-efficacy ($r = 0.40$, $p < 0.001$), psychological ownership ($r = 0.30$, $p < 0.001$), and outcome satisfaction ($r = 0.14$, $p = 0.020$), but was not significantly related to work meaningfulness ($r = 0.01$, $p = 0.941$) or task enjoyment ($r = -0.09$, $p = 0.152$).

To examine how psychological responses might shift in the absence of AI, participants were asked to imagine a scenario in which they typically use AI for work, but on this occasion, AI was unavailable and tasks had to be completed without AI. They then responded to the same psychological measures, now reflecting their anticipated experiences in this no-AI context. The pattern of results remained largely consistent: passive AI reliance was again negatively correlated with self-efficacy ($r = -0.44$, $p < 0.001$), psychological ownership ($r = -0.36$, $p < 0.001$), and outcome satisfaction ($r = -0.25$, $p < 0.001$), while active AI collaboration showed positive correlations with self-efficacy ($r = 0.41$, $p < 0.001$), psychological ownership ($r = 0.32$, $p < 0.001$), and outcome satisfaction ($r = 0.17$, $p = 0.004$). These converging results suggest that the manner in which AI is integrated into work shapes individuals' psychological experiences.

Discussion

Our findings shed light on how generative AI shapes the way people experience their work, and how the manner of AI use matters just as much as its presence. When participants simply copied and pasted AI-generated content, they felt less confident in their own independent abilities, less ownership over the work, and found their tasks less meaningful. This suggests that passive AI use can undermine a sense of competence and agency, likely because the work no longer feels like a personal accomplishment. At the same time, right after using AI, these same participants reported higher task enjoyment and outcome satisfaction compared to other groups. These combined findings paint a nuanced picture that captures some of the mixed implications of using AI:

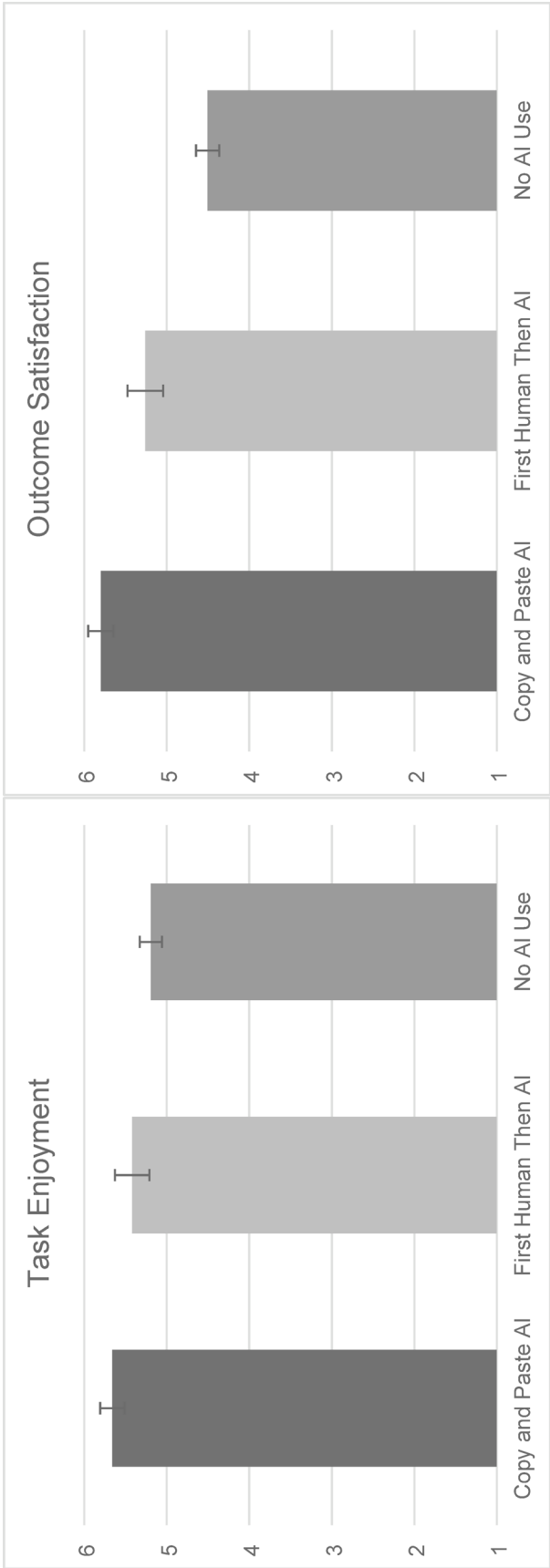


Fig. 3. Secondary Outcomes Following the *Primary Task*. Task enjoyment and outcome satisfaction immediately after participants completed the *Primary Task* across three AI usage conditions: *Copy and Paste AI* (using AI-generated content directly without modification), *First Human Then AI* (writing an initial draft manually and then using AI to refine it), or *No AI Use* (completing the task entirely without AI). Error bars represent SE.

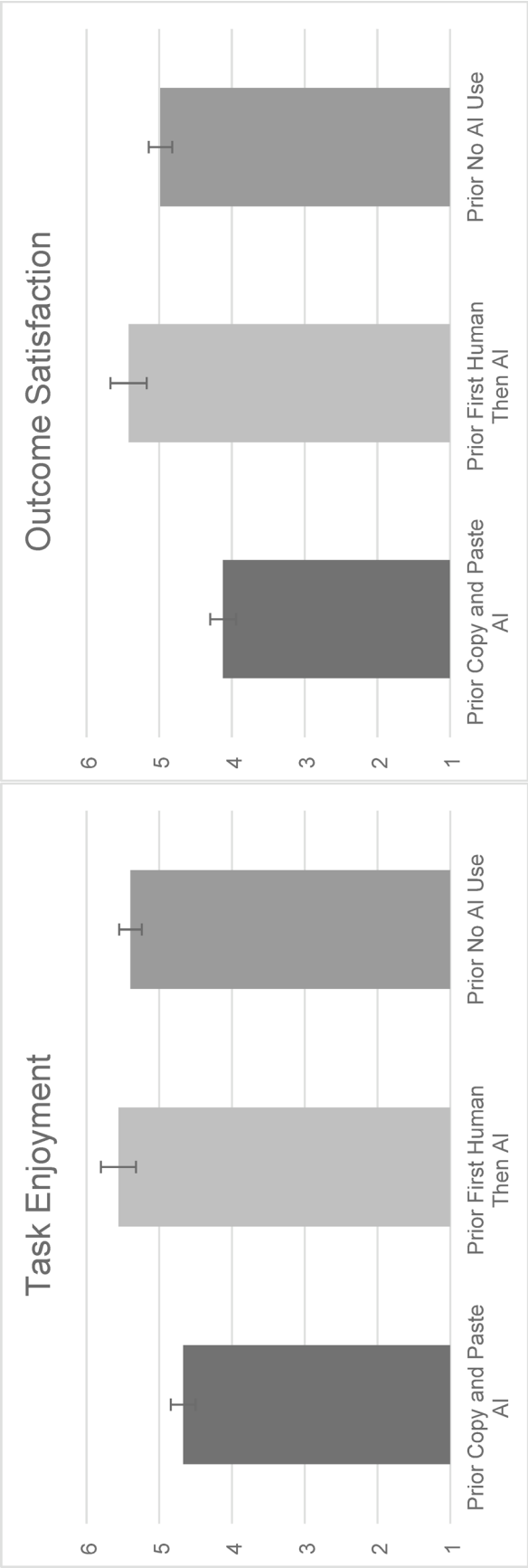


Fig. 4. Secondary Outcomes following the *Subsequent Manual Task*. Task enjoyment and outcome satisfaction after all participants completed the *Subsequent Manual Task* without AI assistance, shown by their prior AI usage condition in the *Primary Task* (*Prior Copy and Paste AI*, *Prior First Human Then AI*, *Prior No AI Use*). Error bars represent SE.

Variable	Mean	SD	1	2	3	4	5	6	7	8	9	10	11	12
1. Passive Reliance	2.68	1.32	–											
2. Active Collaboration	5.62	0.99	–.597**	–										
3. G_Self-Efficacy	6.17	0.88	–.448**	.398**	–									
4. G_Psychological Ownership	5.50	0.78	–.220**	.303**	.375**	–								
5. G_Meaningful Work	5.27	1.36	.052	.005	.056	.089	–							
6. G_Task Enjoyment	4.97	1.49	.091	–.087	–.053	–.011	.762**	–						
7. G_Outcome Satisfaction	5.60	0.96	–.162**	.141*	.294**	.155*	.380**	.434**	–					
8. S_Self-Efficacy	6.21	0.92	–.438**	.411**	.633**	.283**	.052	–.083	.249**	–				
9. S_Psychological Ownership	5.87	0.87	–.359**	.316**	.421**	.326**	.089	–.011	.250**	.454**	–			
10. S_Meaningful Work	5.24	1.39	.014	.020	.061	.036	.857**	.636**	.324**	.085	.144*	–		
11. S_Task Enjoyment	4.62	1.54	–.050	–.018	.073	.015	.629**	.786**	.411**	.123*	.124*	.619**	–	
12. S_Outcome Satisfaction	5.40	1.17	–.248**	.174**	.359**	.129*	.325**	.367**	.633**	.377**	.272**	.316**	.533**	–

Table 1. Descriptive statistics and correlations among key variables. This table presents descriptive statistics for our sample and correlations among our key variables “G” denotes generic psychological measures, while “S” refers to psychological measures assessed after the imaginative, scenario-based task * $p < 0.05$, ** $p < 0.01$

respondents seemed to appreciate that the task was done well and enjoyed the process, but were less likely to see a personal connection to the task, view the task as meaningful, and believe that they could adequately accomplish it on their own.

In contrast, participants who used AI as a collaborator (drafting a version first and then using AI to edit it) didn’t experience these psychological costs. In fact, their self-efficacy and work-related attitudes were statistically similar to those who had not used AI at all. This suggests that when AI is used as a tool to refine one’s own work rather than as a substitute for it, individuals still psychologically “own” the work and retain a sense of competence.

Importantly, many of the key effects persisted even after participants subsequently worked without AI assistance. These lingering effects point to potential longer-term implications of AI usage on professional motivation and identity. Although psychological ownership over work seems to rebound after individuals work on their own without AI assistance, once independent self-efficacy and work meaningfulness are eroded by passive reliance on AI, they may not immediately rebound even when individuals return to independent work. Conceptually, these results accentuate that self-efficacy and work meaningfulness are different from psychological ownership as they represent more internalized beliefs about one’s capabilities and the value of one’s contributions that may be more vulnerable to the use of AI.

Furthermore, whereas after the initial task the group that relied on AI heavily had expressed the highest task enjoyment and outcome satisfaction amongst the three conditions, their enjoyment and satisfaction dropped to the lowest levels amongst the groups when they later worked without AI assistance. This may be a kind of contrast effect: once you have experienced the speed and ease of AI, manual work feels more tedious and less rewarding. By contrast, enjoyment and satisfaction remained stable among those who previously did not use AI or collaborated with AI, suggesting that collaborative or independent work fosters more resilient engagement.

It is important to contextualize these findings within the broader literature showing that generative AI can significantly enhance productivity and performance^{6,40–42}. As generative AI becomes increasingly embedded in professional workflows, avoiding AI altogether may no longer be practical or viable. Indeed, many business leaders now ask employees to maximize their use of generative AI, framing it as a necessary tool for efficiency and competitiveness^{43,44}. Yet this emphasis may inadvertently encourage passive reliance on AI, as doing so saves time and maximizes productivity in the short run. Our findings suggest that extensive reliance on AI harms workers’ psychological connection to their work, eroding AI-independent self-efficacy, and diminishing the sense of meaning that is core to intrinsic motivation. Hence, even if AI integration does not trigger widespread job displacement, its psychological impact on workers remains a critical concern. The central question may thus be not *whether* to integrate AI into work, but *how* to design AI integration that leverages technological capabilities while preserving the human elements essential for psychological flourishing.

Limitations and future directions

An important limitation of the present work is that participants accessed the AI tool outside the survey environment, which constrained our ability to capture fine-grained behavioral interaction data (e.g., timing of AI engagement, prompt-level activity, or stage-specific disengagement). Although we were able to examine attrition at the survey level and capture limited behavioral indicators within the survey itself (e.g., final response word count, time spent on the response submission page, and click count), these measures did not differ across conditions and do not allow precise diagnosis of engagement dynamics or attrition timing. Future studies would benefit from embedding AI interfaces directly within task environments or using instrumented platforms that enable detailed behavioral logging, allowing researchers to link specific interaction patterns to engagement, attrition, and longer-term psychological outcomes.

Relatedly, our operationalization of “active collaboration” focused on a single workflow (First Human Then AI) to provide a clear and enforceable manipulation. In pilot testing, we also examined a closely related First AI

Then Human workflow (i.e., beginning with AI-generated text and then revising it), which produced a similar pattern of results but showed a higher rate of manipulation-check failures. Given the practical challenges of reliably enforcing and verifying this alternative workflow, we focused on one clearer collaboration condition in the main study. Future research should examine a broader range of active collaboration strategies (e.g., using AI for outlining/structuring, more iterative mixed human-AI editing), as different collaboration forms may yield distinct psychological experiences.

In addition, we did not collect baseline measures of AI-related competency or confidence, which could be informative for understanding individual differences in responses to AI-assisted work. While random assignment helps ensure such characteristics are balanced across conditions, future research could examine whether AI-related competence or confidence moderates the psychological consequences of different AI-use workflows.

Understanding and mitigating the psychological risks of AI integration represents one of the most pressing questions for the future of work. Organizations implementing AI should consider training and design interventions that promote active collaboration rather than passive automation.

Materials and methods

This research was reviewed and approved by the University of Southern California Institutional Review Board (UP-24-00,607). All procedures were performed in accordance with relevant guidelines and electronic informed consent was obtained from all participants prior to their participation.

Primary study

We conducted a preregistered online experiment using ChatGPT. A total of 562 individuals entered the study. Of these, 35 participants did not begin the study, and 119 participants began but did not complete it, yielding a final sample of 408 participants recruited through Prolific Academic. Dropout occurred across conditions as follows: *Copy and Paste AI* (25 participants; 21.0%), *First Human Then AI* (48 participants; 40.3%), and *No AI Use* (46 participants; 38.7%). Although dropout rates were somewhat higher in the *First Human Then AI* condition, baseline measures collected prior to the task revealed no significant differences in participants' self-efficacy or self-perceived writing competence across conditions, suggesting that attrition was unlikely to be systematically related to participants' initial confidence or ability. Participants were drawn from a diverse range of occupations and were randomly distributed across experimental conditions to ensure occupational balance, including 68 consultants (16.7%), 89 data analysts (21.8%), 89 human resource professionals (21.8%), 91 managers (22.3%), and 71 marketers (17.4%). Each participant completed two occupation-specific writing tasks adapted from prior research⁶. Each 10-min task was designed to closely mirror real-world tasks in these occupations, such as writing press releases, short reports, analysis plans, and delicate emails.

Following each task, participants completed a set of questionnaires assessing their psychological experiences. In the *Primary Task*, we manipulated the degree of human involvement through random assignment to one of three conditions (*No AI Use*, *Copy and Paste AI*, and *First Human Then AI*). Participants in the *No AI Use* group were asked to complete tasks manually without any AI assistance. Participants in the *Copy and Paste AI* and *First Human Then AI* conditions were instructed to open ChatGPT (i.e., AI assistance) in a separate browser window and use it while completing the task, following condition-specific instructions. Specifically, those in the *Copy and Paste AI* group were asked to use AI-generated content directly without modifications, while those in the *First Human Then AI* group were asked to write the initial draft manually, then use AI assistance to review and edit the draft. This procedure was adapted from prior experimental designs examining generative AI use in writing tasks (e.g.,⁶).

To ensure participants followed their assigned instructions, we took several methodological precautions. These included providing detailed task-specific instructions, embedding a manipulation check question, and disabling the copy-and-paste function for all participants except those in the *Copy and Paste AI* condition. In the *First Human Then AI* condition, participants were required to submit their human-written draft before submitting the AI-edited version. Following the *Primary Task*, all participants completed a manipulation check in which they reported how they actually used (or did not use) AI. Participants were explicitly told that their compensation would not be affected by their response to encourage honest reporting. Of the 408 participants, 269 confirmed that they had followed their assigned AI usage instructions and were included in the main analyses, consistent with our preregistered criteria. Given that a non-negligible number of participants failed the manipulation check, we examined whether these failures reflected potential selection bias; specifically, whether those who failed it systematically differed in their baseline self-efficacy compared to those who passed it. To address this, we compared baseline self-efficacy between the two groups. As reported in the Supplementary Information, mean levels of baseline self-efficacy were similar across both groups, as well as across experimental conditions, suggesting that the failures were not driven by pre-existing differences in participants' perceived ability. Also, as detailed in the Supplementary Information, examination of participant responses suggested that the failures primarily stemmed from comprehension errors or deliberate non-use of AI. Moreover, as a robustness check, we conducted our primary analyses on the full sample, without excluding those who failed. As reported in the Supplementary Information, the results remained largely consistent, indicating that our main findings are robust to these exclusions.

Following the *Primary Task* and associated questionnaires, all participants completed the *Subsequent Manual Task*, which required them to complete another writing assignment manually, without any AI assistance. Task prompts were tailored to participants' occupations and were matched across the two tasks in terms of structure and expected length, allowing for comparisons across time while maintaining role-relevant engagement.

Full details of the design and copies of relevant survey questionnaires are included in the project's OSF page. Descriptive statistics and correlations among key variables appear in Table 2. We performed primary analyses on a final sample of 269 participants (48% female; 94.8% employed; $M_{\text{age}} = 31.85$, $SD_{\text{age}} = 9.86$; 0.4% Native American,

Variable	Mean	SD	1	2	3	4	5	6	7	8
Occupation: HR professional	20.40%									
Occupation: business consultant	17.80%									
Occupation: data analyst	23.00%									
Occupation: manager	19.70%									
Occupation: marketer	19.00%									
Employed	94.80%									
1. Income (\$K)	77.50	4.33	–							
2. Years of tenure in occupation	8.87	8.07	.106	–						
3. Self-efficacy (primary task)	5.42	1.33	.073	.064	–					
4. Psychological ownership (primary task)	4.96	1.14	.047	.067	.208**	–				
5. Meaningful work (primary task)	5.31	1.43	.056	–.122*	.194**	.475**	–			
6. Self-efficacy (manual task)	5.36	1.52	.051	.082	.684**	.170**	.131*	–		
7. Psychological ownership (manual task)	5.53	1.01	.071	.095	.109	.322**	.193**	.204**	–	
8. Meaningful work (manual task)	5.36	1.45	.093	–.156*	.259**	.312**	.666**	.339**	.455**	–

Table 2. Descriptive statistics and correlations among key variables. This table presents descriptive statistics for our sample and correlations among our key variables Employed includes full-time and part-time employment * $p < 0.05$, ** $p < 0.01$

7.8% Asian or Asian American, 46.1% African American, 20.5% Caucasian, 9.7% Hispanic, 15.6% other). Given that a large proportion of our sample identified as African American, we conducted our primary analyses again while controlling for participants' ethnic background. As reported in the Supplementary Information, the results remained largely consistent regardless of whether ethnic background was included as a control.

Measures

Unless otherwise indicated, all measures were evaluated on a 7-point Likert-type response scale, ranging from 1 = strongly disagree to 7 = strongly agree.

Self-efficacy

Participants' self-efficacy was measured using three items adapted from Sherer et al.'s Self-Efficacy Scale (46) ($\alpha = 0.94$): "I feel confident doing similar tasks in the future without the aid of AI tools," "I am certain I can successfully complete tasks like this going forward, without using AI tools," and "I believe I can successfully complete tasks related to my work life without the use of AI assistance."

Psychological ownership

Psychological ownership was assessed using five adapted items from Van Dyne and Pierce⁴⁵'s Psychological Ownership Scale ($\alpha = 0.95$). Example items include: "The task output is MY task output" and "I sense that the output of this task belongs to me."

Meaningful work

Participants' perceived meaningfulness of work was assessed using five items adapted from the Work and Meaning Inventory⁴⁶ ($\alpha = 0.95$). Example items include "I found this type of task meaningful." and "I understand how this type of task would contribute to my life's meaning."

Task enjoyment

We assessed participants' task enjoyment using a single item, "How much did you enjoy doing the writing task?"

Outcome satisfaction

Satisfaction with task output was measured with a single item, "How satisfied are you with the writing you submitted?"

In addition to the above, we collected several exploratory measures at multiple time points in the study. Writing competence and satisfaction with writing quality were assessed prior to the Primary Task (Because baseline writing competence may influence task experiences, we conducted robustness checks that included baseline writing competence [measured prior to the Primary Task] as a covariate. Results were substantively unchanged from those reported in the main text). General self-efficacy (not specifying AI or non-AI use) and affect were assessed prior to the Primary Task, immediately after the Primary Task, and again after the Subsequent Manual Task. AI-dependent self-efficacy was assessed after the Primary Task and after the Subsequent Manual Task. Finally, participants reported their preferred format of AI use (and reasons for their preference) after completing the Subsequent Manual Task. These were not included in the primary analyses reported in this paper to maintain focus on the paper's core theoretical tests and for space considerations. Supplementary analyses of additional self-efficacy measures (general and AI-dependent) are reported in the Supplementary Information. As noted earlier, AI-dependent self-efficacy was preregistered alongside AI-independent self-efficacy (unlike the

other measures that were collected for exploratory purposes); however, we report the AI-dependent self-efficacy analyses in the Supplementary Information.

Follow-up correlational study

To complement the experimental findings, we conducted a separate correlational study using a sample of working adults. A total of 301 participants were initially recruited through CloudResearch Connect. After excluding 16 cases identified on a bot-check question and 15 participants who failed an attention check, the final sample consisted of 270 participants. All participants were based in the United States or the United Kingdom, and provided informed consent prior to beginning the study.

This study aimed to explore how real-world patterns of AI use relate to individuals' psychological experiences at work. Specifically, we distinguished between passive reliance on AI (e.g., using AI-generated content with little to no modification) and active collaboration (e.g., using AI tools to assist, refine, or build upon self-generated work). Participants reported on the extent to which they engaged in each type of AI use in their daily work. They then completed the same core set of dependent variables assessed in the experimental study (i.e., self-efficacy, psychological ownership, meaningful work, task enjoyment, and outcome satisfaction) adapted to reflect participants' general work-related psychological experiences.

In addition, to further examine how individuals anticipate feeling when unable to use AI in their work, participants were also asked to imagine the following scenario: "You typically use AI to do work tasks, but today, AI is unavailable (maybe your Wi-Fi connection is down). You still need to complete the tasks, so you do them without AI. How would you feel in this situation?"

Participants then completed the same psychological measures again, this time based on how they expected to feel in this no-AI context. This design enabled us to compare individuals' general psychological orientations in their current work with their anticipated responses when forced to work without AI tools.

Measures

Unless otherwise indicated, all measures were evaluated on a 7-point Likert-type response scale, ranging from 1 = strongly disagree to 7 = strongly agree.

Generic psychological experiences

Passive reliance

Participants' passive reliance on AI at work was assessed using five items ($\alpha = 0.89$): "I often copy and paste AI-generated content into my work with little to no modification," "I rarely question the accuracy or quality of AI-generated outputs," "I typically do not revise what the AI produces," "When I use AI, I tend to accept its output as-is, even if it differs from how I might approach the task," and "I let AI handle most of the cognitive work in my tasks."

Active collaboration

Participants' active collaboration with AI at work was measured using six items ($\alpha = 0.89$): "I critically evaluate and revise what AI produces," "I treat the AI's output as a draft that I reshape to match my intent," "I often go back and forth with AI to develop better responses," "I treat AI suggestions as helpful input, not final answers," "Even when I incorporate AI's output, I ensure the result reflects my own thinking," and "I use AI primarily to refine my own work rather than to have it create content from scratch."

Self-efficacy, psychological ownership, meaningful work, task enjoyment, and outcome satisfaction

These variables were assessed using similar measures as in the primary (experimental) study, adapted to reflect participants' general psychological experiences in their work life (rather than task-specific reactions): self-efficacy (3 items; $\alpha = 0.91$), psychological ownership (5 items; $\alpha = 0.91$), meaningful work (5 items; $\alpha = 0.95$), task enjoyment (1 item), and outcome satisfaction (1 item).

Scenario-based psychological experiences

Participants completed the same psychological measures as the generic psychological experiences, adapted to reflect their expected experience in the imagined no-AI context: self-efficacy ($\alpha = 0.95$), psychological ownership ($\alpha = 0.93$), meaningful work ($\alpha = .95$), task enjoyment (1 item), and outcome satisfaction (1 item).

Data availability

All data, analysis syntax, preregistration materials, and copies of the survey questionnaires are available on the project's [OSF page](https://osf.io/9qxjz/?view_only=1af9da6203e444fca28c364129274ade).

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Author contributions

E.H.L., Y.Y., N.J., and C.J.W. designed research; E.H.L. performed research; E.H.L. analyzed data; and E.H.L., Y.Y., N.J., and C.J.W. wrote the paper.

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Competing interests

The authors declare no competing interests.

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